

Format. Plots, code, and write-up follow the Homework 1 conventions, and code goes in the same "Markov2026" repository labeled by problem. New this week, normalize histograms as empirical PDFs, state the bin width, and overlay the theory. For acceptance-rejection, also report the empirical acceptance fraction beside $1/c$ and the mean time per accepted sample.

Optional problems. Problems after the END box, and parts marked optional, are not graded.

1. Discrete inverse transform and the cost of a search.

- (a) Let $Z \in \{1, 2, 3\}$ with $P(Z = 1) = 0.05$, $P(Z = 2) = 0.15$, $P(Z = 3) = 0.80$. Give the cumulative thresholds and a three line pseudo-code that returns Z from a single $U \sim U(0, 1)$.
- (b) Count the inequalities checked by your pseudo-code (an "else" is not a comparison). Compute the expected number of comparisons for the order above and for the order that minimizes it. State the rule in one sentence.
- (c) Let X be geometric on $\{1, 2, 3, \dots\}$ with $P(X = k) = p(1-p)^{k-1}$, so that $F(k) = 1 - (1-p)^k$. Show that solving $F(k) \geq U$ for the smallest such k gives

$$X = \left\lceil \frac{\ln(1-U)}{\ln(1-p)} \right\rceil.$$

(a)		Pseudo-code 1	Pseudo-code 2
	$F(z) = \begin{cases} 0.05 & \text{if } z=1 \\ 0.2 & \text{if } z=2 \\ 1 & \text{if } z=3 \end{cases}$	$U = \text{random}(0,1)$ If $(U < 0.05)$ return 1 else if $(U < 0.2)$ return 2 return 3	$U = \text{random}(0,1)$ If $(U < 0.8)$ return 3 else if $(U < 0.95)$ return 2 return 1

(b) 2 inequalities.

Let $N_1(Z=k)$ be the number of comparisons used to return k in Pseudo-code 1.

$$\begin{aligned} E(N_1) &= P(Z=1)N_1(Z=1) + P(Z=2)N_1(Z=2) + P(Z=3)N_1(Z=3) \\ &= 0.05 \cdot 1 + 0.15 \cdot 2 + 0.8 \cdot 2 \\ &= 1.95 \end{aligned}$$

Let $N_2(Z=k)$ be the number of comparisons used to return k in Pseudo-code 2.

$$\begin{aligned} E(N_2) &= P(Z=1)N_2(Z=1) + P(Z=2)N_2(Z=2) + P(Z=3)N_2(Z=3) \\ &= 0.8 \cdot 1 + 0.15 \cdot 2 + 0.05 \cdot 2 \\ &= 1.2 \end{aligned}$$

Assign probabilities in descending order to minimize running deeper into control statements.

(c) $u \leq F(k) = 1 - (1-p)^k$ where $k \geq 1$.

$$-u \geq (1-p)^k - 1$$

$$1 - u \geq (1-p)^k \geq 0 \quad \text{since } 1-p \geq 0.$$

Since $\ln$ is monotonically increasing over its domain of positive reals,

$$\ln(1-u) \geq \ln[(1-p)^k]$$

$$\ln(1-u) \geq k \ln(1-p)$$

$$\frac{\ln(1-u)}{\ln(1-p)} \leq k \quad \text{ineq flips since } 1-p < 1 \text{ and } \ln(1-p) < \ln(1) = 0.$$

Since k is an integer, $\left\lceil \frac{\ln(1-u)}{\ln(1-p)} \right\rceil \leq k$ and

$X = \left\lceil \frac{\ln(1-u)}{\ln(1-p)} \right\rceil$ is the smallest positive integer satisfying the original inequality.

2. Inverse transform on the whole line, and a law of large numbers that fails. Let X have the standard Cauchy density $f(x) = \frac{1}{\pi(1+x^2)}$, $x \in \mathbb{R}$.

(a) Derive F and F^{-1} , and give two line pseudo-code for sampling X .

(b) Show that the median of $|X|$ is 1 while $E|X| = \infty$. Explain in one or two sentences how a typical draw can be of order one while the mean fails to exist.

$$\begin{aligned}
 (a) \quad F(x) &= \int_{-\infty}^x \frac{1}{\pi(1+s^2)} ds = \lim_{t \rightarrow -\infty} \int_t^x \frac{1}{\pi(1+s^2)} ds = \lim_{t \rightarrow -\infty} \frac{1}{\pi} [\arctan(x) - \arctan(t)] \\
 &= \frac{1}{\pi} \left[\arctan(x) + \frac{\pi}{2} \right] = \boxed{\frac{1}{\pi} \arctan(x) + \frac{1}{2}}
 \end{aligned}$$

$$y = \frac{1}{\pi} \arctan(x) + \frac{1}{2}$$

$$x = \frac{1}{\pi} \arctan(y) + \frac{1}{2}$$

Pseudo-code for sampling X

$U = \text{random}(0, 1)$

return $\tan(\pi(U - 0.5))$

$$\pi\left(x - \frac{1}{2}\right) = \arctan(y)$$

$$\tan\left(\pi\left(x - \frac{1}{2}\right)\right) = y$$

$$\boxed{F^{-1}(x) = \tan\left(\pi\left(x - \frac{1}{2}\right)\right)}, \quad 0 < x < 1.$$

$$(b) \quad P(|X| \leq t) = P(-t \leq X \leq t) = \int_{-t}^t \frac{1}{\pi(1+s^2)} ds$$

$$= \frac{1}{\pi} [\arctan(t) - \underbrace{\arctan(-t)}_{\text{odd fn}}]$$

$$= \frac{2}{\pi} \arctan(t)$$

$$\text{median: } \frac{1}{2} = P(|X| \leq t) = \frac{2}{\pi} \arctan(t)$$

$$\frac{\pi}{4} = \arctan(t)$$

$$t = 1 \quad \boxed{\text{median of } |X| \text{ is } 1.}$$

$$E(|X|) = \int_{-\infty}^{\infty} |x| \frac{1}{\pi(1+x^2)} dx = \underbrace{\int_{-\infty}^0 -x \frac{1}{\pi(1+x^2)} dx}_{(1)} + \underbrace{\int_0^{\infty} x \frac{1}{\pi(1+x^2)} dx}_{(2)}$$

$$\textcircled{1} \quad u = 1+x^2 \\ du = 2x dx$$

$$\int_{u=\infty}^{u=1} -\frac{1}{2} \frac{du}{\pi u} = \left[-\frac{1}{2\pi} \ln|u| \right]_{\infty}^1 = \lim_{t \rightarrow \infty} \left\{ -\frac{1}{2\pi} \ln|1| + \frac{1}{2\pi} \ln|t| \right\} = \infty$$

$$\textcircled{2} \quad \int_0^{\infty} x \frac{1}{\pi(1+x^2)} dx = \int_1^{\infty} \frac{1}{2\pi} \frac{du}{u} = \lim_{s \rightarrow \infty} \frac{1}{2\pi} \{ \ln|s| - \ln|1| \} = \infty$$

Hence $E(|X|)$ diverges to ∞ .

Although most draws are near 1,
mean can fail to exist if probability doesn't decay fast enough
to prevent rare but large draws from contributing "enough"
to the expected value.

3. **Acceptance-rejection with an exponential envelope.** Let $X \sim \text{Gamma}(2, 1)$ with density $f(x) = xe^{-x}$ for $x \geq 0$. The CDF is $F(x) = 1 - (1+x)e^{-x}$, which cannot be inverted in closed form, so we use rejection with the proposal $g(x) = \lambda e^{-\lambda x}$.

- Take $\lambda = 1/2$. Show that the smallest valid envelope constant is $c = 4/e$, state the theoretical acceptance rate, and show that the acceptance test simplifies to $U < \frac{e}{2} X e^{-X/2}$.
- Give succinct pseudo-code (single while loop), sampling the proposal by inversion.
- Implement this for $N = 10^4$ accepted samples. Plot the normalized histogram with $f(x)$ overlaid, report the empirical acceptance fraction against $1/c$, and report the mean time per accepted sample. Then repeat with $\lambda = 0.2$, for which $c = 1/(0.16e) = 2.2993$, and comment on the difference.

$$(a) \quad \frac{f(x)}{c g(x)} \leq 1$$

$$\frac{f(x)}{g(x)} \leq c$$

$$\frac{f(x)}{g(x)} = \frac{x e^{-x}}{\lambda e^{-\lambda x}} = \frac{x}{\lambda} e^{x(\lambda-1)}$$

$$\text{minimize } \frac{f}{g} : \left(\frac{f(x)}{g(x)} \right)' = \frac{1}{\lambda} e^{x(\lambda-1)} + \frac{x}{\lambda} (\lambda-1) e^{x(\lambda-1)}$$

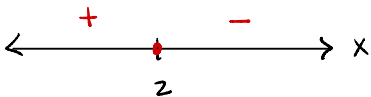
$$\text{critical pts: } 0 = e^{x(\lambda-1)} \frac{1}{\lambda} (1 + x(\lambda-1))$$

$$0 = 1 + x(\lambda-1)$$

$$\frac{-1}{\lambda-1} = x$$

$$x = \frac{1}{1-\lambda} = \frac{1}{1-1/2} = 2$$

$$\left(\frac{f}{g} \right)' = 2e^{-x/2} (1 - x/2)$$



$x=2$ is global max for $\frac{f}{g}$.

$$\frac{f(2)}{g(2)} = 2 \cdot 2 \cdot e^{2(-1/2)} = \frac{4}{e}$$

$$\frac{f(x)}{c g(x)} \leq 1 \Rightarrow \frac{f(x)}{g(x)} \leq c \Rightarrow \frac{4}{e} \leq c.$$

The smallest value for c is $4/e$. $P(\text{accept}) = \frac{1}{c} = \frac{e}{4} \approx 0.679570$

acceptance test: $u < \frac{p_Y}{c q_Y} = \frac{e}{4} 2Y e^{-Y/2}$

$$u < \frac{e}{2} Y e^{-Y/2}$$

(b) proposal density: $g(x) = \frac{1}{2} e^{-x/2}$

compute cdf: $G(x) = \int_0^x \frac{1}{2} e^{-t/2} dt = [-e^{-t/2}]_0^x = -e^{-x/2} + 1$

$$= 1 - e^{-x/2}$$

Inverse cdf: $V = 1 - e^{-Y/2}$

$$e^{-Y/2} = 1 - V$$

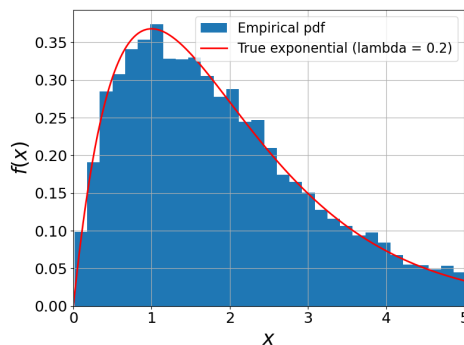
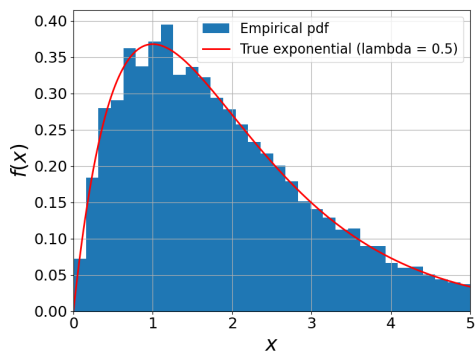
$$-\frac{Y}{2} = \ln(1 - V)$$

$$Y = -2 \ln(1 - V)$$

Pseudo-code for a single acceptance

```
x = -1
c = 4/e
lm = 0.5
while (x < 0)
  v = random(0,1)
  y = -2*ln(1-v)
  u = random(0,1)
  if (u < y/(c*lm) * exp(y*(lm-1)))
    x = y
```

(c) Smaller lambda led to decreased efficiency as the acceptance rate dropped by about 25%. Both rates are close to their theoretical values of $1/c$. The mean time for acceptance almost doubled going from $\lambda = 0.5$ to 0.2 .



```
(base) junhong@MacBook-Pro-2 hw2 % python p3c-1.py
Acceptance rate: 0.6776
1/c: 0.6796
Mean time per accepted sample: 0.000003 seconds
```

```
• (base) junhong@MacBook-Pro-2 hw2 % python p3c-2.py
Acceptance rate: 0.4283
1/c: 0.4349
Mean time per accepted sample: 0.000005 seconds
```

4. **Ion channel dwell times and mixture sampling.** A patch clamp record of a single channel is a sequence of closed and open intervals. Markov gating models predict that dwell times in a set of states are mixtures of exponentials, with one component per state in the set (Colquhoun and Hawkes, *Phil. Trans. R. Soc. Lond. B* 1982). Take a channel with a brief and a long closed state, so that the closed dwell time T has density

$$f(t) = a\lambda_f e^{-\lambda_f t} + (1-a)\lambda_s e^{-\lambda_s t}, \quad t \geq 0,$$

with $a = 0.9$, $\lambda_f = 1000 \text{ s}^{-1}$ and $\lambda_s = 10 \text{ s}^{-1}$.

- (a) Verify that f is a density and compute $E[T]$, $\text{Var}(T)$, and the coefficient of variation $\text{CV} = \sqrt{\text{Var}(T)}/E[T]$. Compare the CV with that of a single exponential and say in one or two sentences what the comparison tells you.
- (b) Give a three step composition algorithm using one Bernoulli draw and one inverse transform draw, and show by conditioning on the component that it returns a sample from f .
- (c) Generate 10^5 dwell times by composition. Plot the normalized histogram on semilog vertical axes with $f(t)$ overlaid, so that both components are visible. Report the empirical mean and the empirical $P(T > 50 \text{ ms})$ against theory.

$$\begin{aligned} \text{(a)} \quad \int_0^\infty f(t) dt &= \int_0^\infty (0.9 \cdot 1000 \cdot e^{-1000t} + 0.1 \cdot 10 e^{-10t}) dt \\ &= \int_0^\infty (900 e^{-1000t} + e^{-10t}) dt = 900 \int_0^\infty e^{-1000t} dt + \int_0^\infty e^{-10t} dt \\ &= \frac{900}{-1000} [e^{-1000t}]_0^\infty + \frac{1}{-10} [e^{-10t}]_0^\infty \\ &= -\frac{9}{10} [0 - 1] - \frac{1}{10} [0 - 1] = 1 \quad \checkmark \end{aligned}$$

$$\begin{aligned} E(T) &= \int_0^\infty t f(t) dt = \int_0^\infty t (0.9 \cdot 1000 \cdot e^{-1000t} + 0.1 \cdot 10 e^{-10t}) dt \\ &= \lim_{s \rightarrow 1} \left\{ 0.9 \frac{d}{ds} \left[\int_0^\infty e^{-1000ts} dt \right] + 0.1 \frac{d}{ds} \left[\int_0^\infty e^{-10ts} dt \right] \right\}, \quad s > 0 \\ &\quad \left[\frac{1}{1000s} e^{-1000ts} \right]_{t=0}^{t=\infty} = \frac{-1}{1000s} \quad \left[\frac{1}{10s} e^{-10ts} \right]_{t=0}^{t=\infty} = \frac{-1}{10s} \end{aligned}$$

$$= \lim_{s \rightarrow 1} 0.9 \frac{1}{1000s^2} + 0.1 \frac{1}{10s^2} = \frac{0.9}{1000} + \frac{0.1}{10} = 0.0109$$

$$E(T) = 0.0109 \text{ s}$$

$$E(T^2) = \int_0^{\infty} t^2 f(t) dt = \int_0^{\infty} t^2 (0.9 \cdot 1000 \cdot e^{-1000t} + 0.1 \cdot 10 \cdot e^{-10t}) dt$$

$$= \lim_{s \rightarrow 1} \left\{ \underbrace{\frac{0.9}{1000} \frac{d^2}{ds^2} \left[\int_0^{\infty} e^{-1000ts} dt \right]}_{\frac{1}{1000s}} + \underbrace{\frac{0.1}{10} \frac{d^2}{ds^2} \left[\int_0^{\infty} e^{-10ts} dt \right]}_{\frac{1}{10s}} \right\}, \quad s > 0$$

$$= \lim_{s \rightarrow 1} \frac{0.9}{1000^2} \frac{2}{s^3} + \frac{0.1}{10^2} \frac{2}{s^3} = \frac{1.8}{1000^2} + \frac{0.2}{100} = 2.0018 \times 10^{-3} \text{ s}^2$$

$$\text{Var}(T) = E(T^2) - E(T)^2$$

$$= [2.0018 \times 10^{-3} - (0.0109)^2] \text{ s}^2$$

$$\text{Var}(T) = 0.00188299 \text{ s}^2$$

$$CV = \frac{\sqrt{\text{Var}(T)}}{E(T)} = \frac{\sqrt{0.00188299}}{0.0109} \approx 3.981049$$

CV for a single exponential is $CV = \frac{\sqrt{\lambda^2}}{\lambda} = 1$.
 Larger CV shows that long closures contribute significantly to the spread of the distribution despite short closures being more common.

(b) Step 1: Generate a random number U against "a" to determine if the channel is brief or long closed.

Step 2: Set rate to λ_{fast} or λ_{slow} accordingly.

Step 3: Generate a random number V and set $T = -\ln(1-V)/\text{rate}$

exponential cdf

$$P(T \leq t) = P(\text{fast}) P(T \leq t | \text{fast}) + P(\text{slow}) P(T \leq t | \text{slow})$$

$$= a(1 - e^{-\lambda_f t}) + (1-a)(1 - e^{-\lambda_s t})$$

$$f(t) = \frac{d}{dt} P(T \leq t) = a(-\lambda_f)(-e^{-\lambda_f t}) + (1-a)(-\lambda_s)(-e^{-\lambda_s t})$$

$$= a\lambda_f e^{-\lambda_f t} + (1-a)\lambda_s e^{-\lambda_s t} \quad \checkmark$$

5. **Simulation Experiment:** growth by redirection, and an average nobody obeys. Hartle, Krapivsky, Redner and Zhang (*arXiv:2604.01540*, 2026) study trees grown by isotropic redirection, in which each new node picks an existing node uniformly at random and then attaches to a randomly chosen neighbor of that node. Almost every node ends up a leaf, and the number C of core nodes (degree at least two) grows sublinearly, $C \sim N^\mu$ with $\mu \approx 0.566$. Their appendix strips the model down to a two population caricature that tracks only the core count,

$$C(N+1) = \begin{cases} C(N) + 1, & \text{with probability } \mu C(N)/N, \\ C(N), & \text{otherwise,} \end{cases} \quad C(3) = 1,$$

with $0 < \mu \leq 1$. Each step is one Bernoulli draw, so this is a sampling problem. Take $\mu = 1$ throughout, the marginal case, for which one can show that $z = C(N)/N$ has the limiting density

$$h(z) = 2(1-z), \quad 0 < z < 1.$$

- Check that h is a density and compute $E[z]$, $\text{Var}(z)$, and the ratio of standard deviation to mean.
- Sample h by inversion. Derive H and H^{-1} , give two line pseudo-code, and generate 10^5 samples. Check the sample mean against part (a).
- Simulate the growth process at $\mu = 1$ out to $N = 10^4$ for $R = 10^3$ independent realizations, vectorizing across realizations. By this I mean that the update at step n must draw a single length R vector of Bernoulli variates: a loop over realizations wrapped around a loop over steps is 10^7 interpreted iterations and will not finish in a sitting. Plot the normalized histogram of $z = C/N$ with $h(z)$ overlaid, and report the empirical mean and the empirical ratio of standard deviation to mean against your answers in part (a). Report the smallest and largest core you saw, and comment in two or three sentences on what the average core size does and does not tell you about a single network.

$$(a) \int_0^1 h(z) dz = \int_0^1 2(1-z) dz = \left[-2 \frac{(1-z)^2}{2} \right]_0^1 = 0 + 1 = 1 \quad \checkmark$$

$$E(z) = \int_0^1 z 2(1-z) dz = 2 \int_0^1 (z - z^2) dz = 2 \left[\frac{z^2}{2} - \frac{z^3}{3} \right]_0^1 = 2 \left[\frac{1}{2} - \frac{1}{3} \right] = \frac{1}{3}$$

$$E(z^2) = \int_0^1 z^2 2(1-z) dz = 2 \int_0^1 (z^2 - z^3) dz = 2 \left(\frac{1}{3} - \frac{1}{4} \right) = \frac{1}{6}$$

$$\text{Var}(z) = E(z^2) - E(z)^2 = \frac{1}{6} - \left(\frac{1}{3} \right)^2 = \frac{1}{6} - \frac{1}{9} = \frac{1}{18}$$

$$CV = \frac{\sqrt{\text{Var}(z)}}{E(z)} = \frac{\sqrt{1/18}}{1/3} = \frac{1}{\sqrt{2}}$$

$$(b) H(z) = \int_0^z h(t) dt = \int_0^z 2(1-t) dt = \left[2(t - t^2/2) \right]_0^z = 2(z - z^2/2) \\ = 2z - z^2 = 1 - (z-1)^2, \quad 0 < z < 1$$

Inverse

$$\begin{aligned} z &= 1 - (y-1)^2 \\ (y-1)^2 &= 1-z \\ y-1 &= \pm \sqrt{1-z} \\ y &= 1 \pm \sqrt{1-z} \end{aligned}$$

we require $0 \leq y \leq 1$.

Hence

$$H^{-1}(z) = 1 - \sqrt{1-z}$$